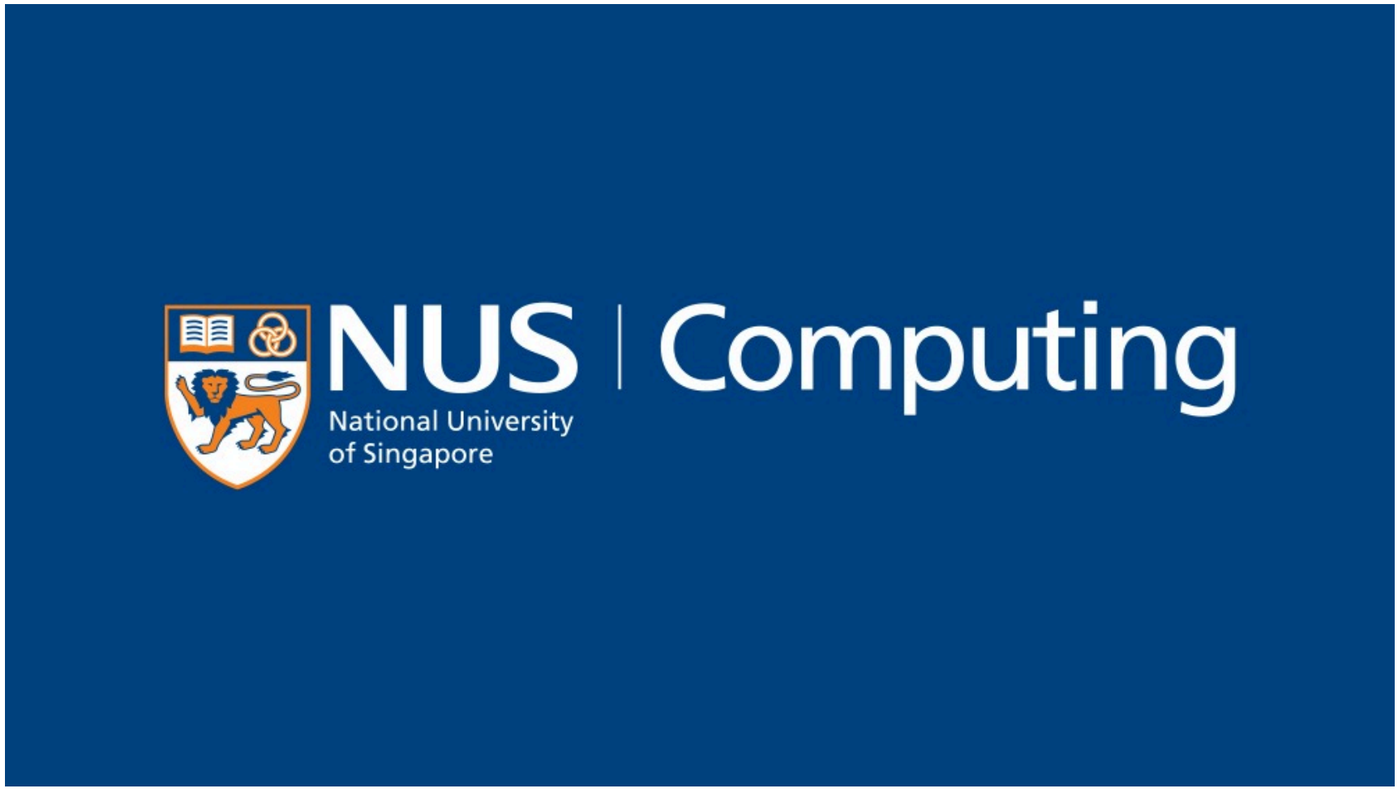
Recent Announcements

Masterclass on Generative AI with Arpit Jain Recording Available
(https://classroom.emeritus.org/courses/15754/discussion_topics/1135443)
Dear Participant, The recording for Masterclass on Generative AI with Arpit Jain conducted on Wednesday, October 21, 2026, at 08…
Posted on: Oct 22, 2026, 7:31 AM

2 hour Reminder: Masterclass on Generative AI with Arpit Jain
(https://classroom.emeritus.org/courses/15754/discussion_topics/850323)
Dear Participant, This is a 2 hour reminder for an upcoming live session: Presenter(s):· Arpit Jain Session Title: Masterclass on Gen…
Posted on: Oct 21, 2026, 6:00 PM

24 hour Reminder: Masterclass on Generative AI with Arpit Jain
(https://classroom.emeritus.org/courses/15754/discussion_topics/850322)
Dear Participant, This is a 24 hour reminder for an upcoming live session: Presenter(s):· Arpit Jain Session Title: Masterclass on Ge…
Posted on: Oct 20, 2026, 8:00 PM

Start Here (https://classroom.
Programme Outline (https://classroom.
Completion Requirements (https://classroom.em
Meet the Faculty (https://cla



Welcome to NUS Advanced Computing for Executives, School of Computing's **AI, ML and Data Science** programme!

Dear Participant,

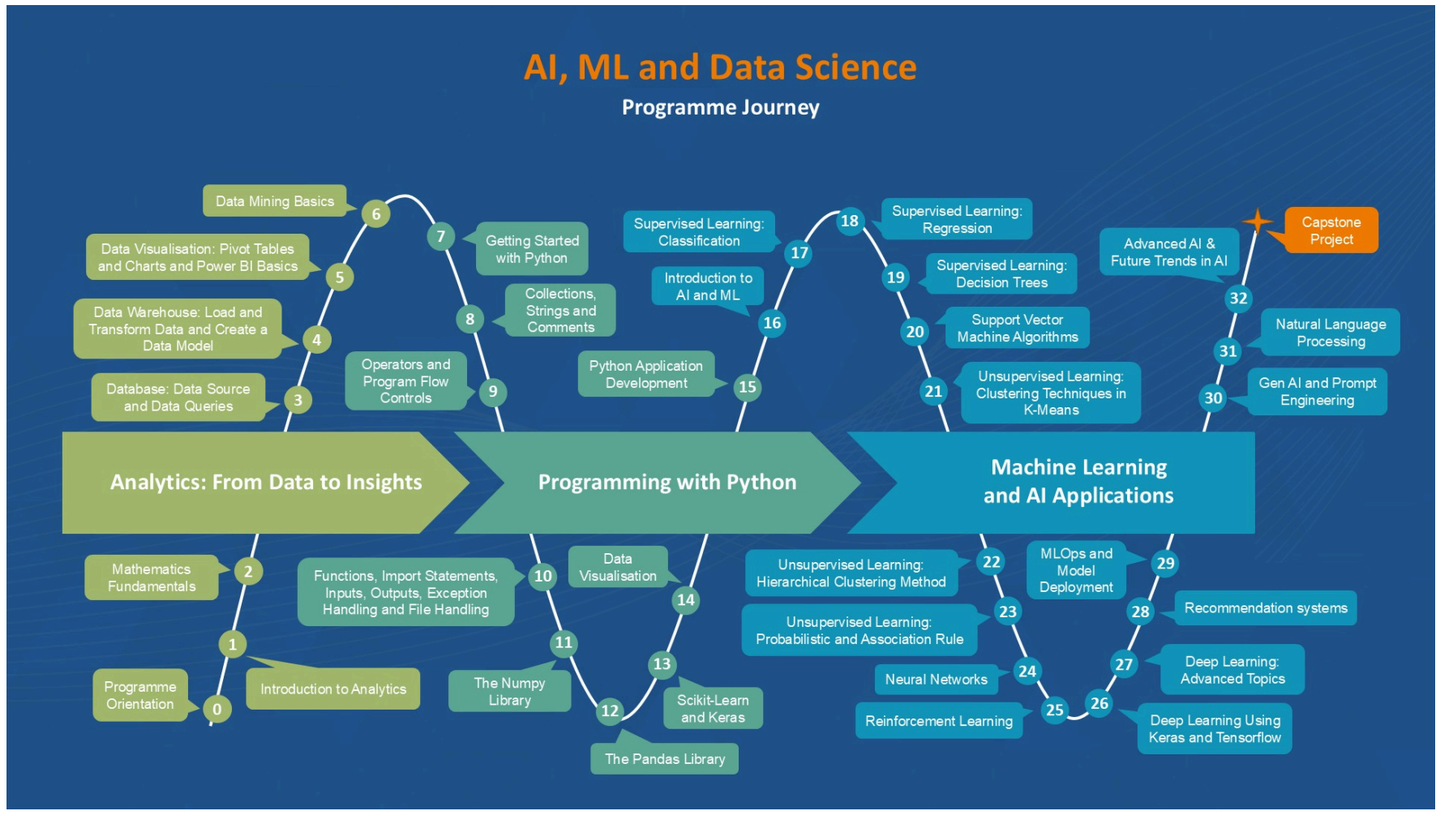
Organisations today generate and rely on unprecedented volumes of data to inform decisions, streamline operations, and spark innovation. This programme brings together the essential capabilities of **data analytics, Python programming, machine learning (ML) and artificial intelligence (AI)** to help you turn raw data into strategic advantage for your organisation.

We begin by grounding you in **data analytics**, focusing on how to process, clean, visualise, and interpret data to discover patterns, reveal opportunities, and drive evidence-based decisions. You will gain hands-on experience with data analytics tools, building confidence in your ability to extract insights from a variety of data sources.

Next, you'll advance into **Python programming,** one of the most widely used languages for data science and AI. Using libraries like NumPy, Pandas, Scikit-learn, TensorFlow, and Keras, you will learn to automate data workflows, build models and create scalable solutions that go far beyond standard analytics.

In the **ML and AI** sections, you'll explore supervised and unsupervised learning, neural networks, deep learning, reinforcement learning, recommendation engines, and generative AI applications. With tools such as Hugging Face and OpenAI Playground, you'll see how cutting-edge AI systems are built, deployed and optimised in real-world contexts.

Here is a quick look at the roadmap of the programme elements:



By the end of this programme, you will be able to:

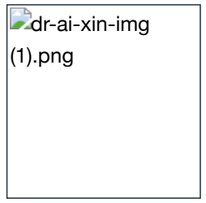
- **Data-Driven Decision-Making with Statistical and Analytical Techniques:**
  Gain foundational knowledge in statistics and analytics to extract meaningful insights from raw data and support strategic business decisions.
- **Database Management and Modelling with DB Browser for SQLite, Power BI Desktop and Excel:**
  Build, clean and manipulate data using industry standard tools. Create relational databases, construct data models and prepare datasets for visualisation and analysis.

- **Python for Data Science: End-to-End Data Workflows:**
  Use Python with core libraries like NumPy, Pandas and Scikit-Learn to automate data collection, cleaning, transformation, analysis and modelling for scalable solutions.
- **Applied Machine Learning: Algorithms and Use Cases:**
  Implement supervised and unsupervised machine learning algorithms including classification, regression, clustering and association rule mining. Understand model evaluation and optimisation for real-world problems.
- **Deep Learning with TensorFlow and Keras:**
  Design, train and optimise deep learning architectures such as neural networks and CNNs. Use TensorFlow and Keras to solve complex tasks like image recognition, sentiment analysis and more.
- **Operationalising AI: Machine Learning Operations and Model Lifecycle Management:**
  Deploy and manage AI models at scale using Machine Learning Operations principles. Leverage tools like MLflow and version control to track experiments, manage reproducibility and streamline model updates and deployment.
- **Recommendation Engines and Reinforcement Learning:**
  Build and evaluate recommendation systems using collaborative filtering, content-based filtering and hybrid approaches. Explore reinforcement learning to develop models that learn from dynamic environments and feedback loops.
- **Generative AI and Natural Language Processing (NLP):**
  Explore the power of generative AI and NLP with tools such as OpenAI Playground, Hugging Face and Transformer-based models. Apply these technologies to a wide range of language and AI applications.
- **Responsible AI - Explainability, Fairness and Governance:**
  Ensure transparency, fairness and accountability in AI models. Apply responsible AI principles to enhance explainability, mitigate bias and uphold ethical standards in AI deployment.

The **AI, ML and Data Science** programme requires a weekly time commitment of six to eight hours. Each module will be made available on a per-week basis. The programme spans 33 weeks comprised of 32 weeks of programme work and 1 week to complete the capstone project. Throughout the programme, concepts are reinforced through examples, live masterclasses, peer discussions and case studies, ensuring every skill you gain is practical and directly applicable to your role. Your Programme Leader will provide the guidance you need to comprehend the material.

Your programme commences on **December 23, 2025.**

We wish you the best in this learning journey.

 **Dr Ai Xin** Lecturer NUS Advanced Computing for Executives, School of Computing	 **Dr Danny Poo** Associate Professor NUS Advanced Computing for Executives, School of Computing	 **Dr Amirhassan Monadjemi** Senior Lecturer NUS Advanced Computing for Executives, School of Computing

Programme Support Info

The programme begins with a self-paced Orientation Week to allow you to familiarise yourself with the learning platform and to introduce yourself to other participants. Coursework, including any live events, starts after the Orientation Week. You will receive a notification about the programme start on or shortly before the start date listed below.

Key Dates

Start Date: Tuesday, December 23, 2025
End Date: Friday, September 18, 2026
Programme access until: Saturday, September 18, 2027

Support



For assistance, you can reach out to the Programme Support team at programsupport@emeritus.org (mailto:programme.support@emeritus.org)
You may expect a response within 24-48 hours, Monday to Saturday.

Mobile Access



If you want to access Canvas on mobile, you can use the mobile browser and access the course using the URL - https://classroom.emeritus.org (https://classroom.emeritus.org/)

Grievances



At any point, if you feel that your query could have been answered in a better way, please write to us at - escalation.apac@emeritus.org (mailto:escalation.apac@emeritus.org) for our leadership to investigate further.

Programme Policy

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